Sample Selection and Missing Data

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1 Introduction

Introduction

- Consider a wealth equation.

$$wealth = \beta_0 + \beta_1 pension + \beta_2 educ + \beta_3 age + \beta_4 income + u.$$

  The population is all families in the US.

- Assuming u has zero mean and is uncorrelated with $x \equiv (1, pension, educ, age, income)'$, OLS will return the consistent estimates of β provided that we have a *random sample from the population.*

Sampling based on exogenous explanatory variable

- Suppose the sample contains only families whose household head is less than 65 years old ($age < 65$). Would OLS with the *selected sample* be consistent for β?

- It will be shown that if we have

$$\mathrm{E}(u \mid x) = \mathrm{E}(u \mid pension, educ, age, income) = 0,$$

  OLS is still consistent for β.

- Recall with random sample, we require $\mathrm{Corr}(u, x) = 0$, which is weaker than the above mean independence assumption.

Sampling based on a response variable

- Suppose only families with positive wealth are included in the sample.
- Unfortunately, it turns OLS in this case will be inconsistent even under the previous mean independence assumption.

Incidental truncation and self-selection

- Sometimes the sample selection is not a deterministic function of either x_j or y as above, but it may be related to them.
- This includes the missing data case, where data are missing on (x, y) for some units drawn randomly from the population. We don't have a rule to claim that certain (x_i, y_i) will be missing.
- Another example is when y is observed only when a certain event is true. For example, in the estimation of wage equation, the dependent variable is $\log(wage^o)$, where $wage^o$ is "wage offer". We can observe one's $wage^o$ only when the person decides to work.
- This problem is generally called "*incidental truncation*".
- *Self-selection* is the most common reason for incidental truncation. For example, we observe wage offer only if the person "self-selects" into the workforce.

2 When can we ignore sample selection?

Linear model

- Consider linear model with (possibly) endogenous explanatory variables:

$$y = \beta_1 + \beta_2 x_2 + \cdots + \beta_K x_K + u = x'\beta + u, \quad \mathrm{E}(zu) = 0.$$

  Let $x = (1, x_2, \ldots, x_K)'$. z is a $L \times 1$ vector of instrumental variables. If x are all exogenous, $z = x$.

- Given a random sample from the population, we can use 2SLS to consistently estimate β under a rank condition, which is essentially $\operatorname{rank} \mathrm{E}(zx') = K$.

- Unfortunately, the rank condition and $\mathrm{E}(zu) = 0$ (no correlation) are not usually enough to consistently estimate β with *a selected sample*.

- Consider a special case $z = x$ first.

- To analyze this problem, it is helpful to think of drawing units randomly from the population, but now the random draw for unit i, (x_i, y_i, z_i), is supplemented by drawing a *selection indicator*, $s_i \in \{0, 1\}$. We observe unit i, hence (x_i, y_i, z_i), if and only if $s_i = 1$.

- Therefore, our "data" are $\{(x_i, y_i, z_i, s_i) : i = 1, \ldots, n\}$, and s_i determines the actual sample for estimation.

- For simplicity, consider the just identification case, $\dim(z_i) = \dim(x_i)$.

- The IV estimator using the selected sample:

$$\begin{aligned}\hat{\beta}_{IV} &= \left(\sum_{i=1}^{n} s_i z_i x_i'\right)^{-1} \left(\sum_{i=1}^{n} s_i z_i y_i\right) \\ &= \beta + \left(n^{-1} \sum_{i=1}^{n} s_i z_i x_i\right)^{-1} \left(n^{-1} \sum_{i=1}^{n} s_i z_i u_i\right).\end{aligned}$$

  The trick is if i was not selected, $s_i = 0$.

- By the LLN for random samples,

$$\operatorname{plim}_{n\to\infty} \hat{\beta}_{IV} = \beta + (\mathrm{E}(szx'))^{-1} \mathrm{E}(szu).$$

- Sufficient conditions for consistency using selected sample are

$$\operatorname{rank} \mathrm{E}(szx') = K \quad \text{and} \quad \mathrm{E}(szu) = 0.$$

- Note

$$\mathrm{E}(szx') = \mathrm{E}(s\,\mathrm{E}(zx' \mid s)) = \Pr(s = 1) \cdot \mathrm{E}(zx' \mid s = 1).$$

  The rank condition is equivalent to

$$\operatorname{rank} \mathrm{E}(zx' \mid s = 1) = K.$$

- This means for the selected sample we should have enough number of IV. If x are all exogenous, there should not be perfect multicollinearity in the selected sample.

- The general 2SLS estimator is

$$\begin{aligned}\hat{\beta}_{2SLS} &= A_s^{-1}\left[\left(n^{-1}\sum_{i=1}^{n} s_i z_i x_i'\right)'\left(n^{-1}\sum_{i=1}^{n} s_i z_i z_i'\right)^{-1}\left(n^{-1}\sum_{i=1}^{n} s_i z_i y_i\right)\right] \\ &= \beta + A_s^{-1}\left(n^{-1}\sum_{i=1}^{n} s_i z_i x_i'\right)'\left(n^{-1}\sum_{i=1}^{n} s_i z_i z_i'\right)^{-1}\left(n^{-1}\sum_{i=1}^{n} s_i z_i u_i\right).\end{aligned}$$

  where $A_s = \left(n^{-1}\sum_{i=1}^{n} s_i z_i x_i'\right)'\left(n^{-1}\sum_{i=1}^{n} s_i z_i z_i'\right)^{-1}\left(n^{-1}\sum_{i=1}^{n} s_i z_i x_i'\right)$.

- For the general 2SLS case, the conditions (excepting for the regularity conditions) for consistency of OLS are

$$\mathrm{E}(szu) = 0 \quad \operatorname{rank}\mathrm{E}(zz' \mid s=1) = L \quad \operatorname{rank}\mathrm{E}(zx' \mid s=1) = K.$$

- The rank condition usually holds and is testable.

- $\mathrm{E}(szu) = 0$ is more interesting.

Missing at random

- **Key point**: to make 2SLS or OLS consistent, we need $\mathrm{E}(szu) = 0$ or $\mathrm{E}(sxu) = 0$ if $z = x$.

- If $s \perp\!\!\!\perp (z, u)$ and $\mathrm{E}(zu) = 0$,

$$\mathrm{E}(szu) = \mathrm{E}(zu) = 0.$$

- If $s \perp\!\!\!\perp (z, u)$ equivalently $s \perp\!\!\!\perp (z, x, y)$, the data are called "*missing at random*".

Exogenous sampling

- The following condition is more practically relevant

$$\mathrm{E}(u \mid z, s) = \mathrm{E}(u \mid z) = 0. \tag{1}$$

  Then

$$\mathrm{E}(szu) = \mathrm{E}(sz\,\mathrm{E}(u \mid s, z)) = 0.$$

- This condition has two parts:

  1. $\mathrm{E}(u \mid z, s) = \mathrm{E}(u \mid z)$ says u and s are mean independent given z.
  2. $\mathrm{E}(u \mid z) = 0$, which is stronger than exogeneity requirement on the instruments, which only needs $\mathrm{Corr}(u, z) = 0$.

- If $\mathrm{E}(u \mid z, s) = \mathrm{E}(u \mid z)$, we call it *exogenous sampling*.

- If s is a *deterministic* function of z, say $s = h(z)$. The above condition holds of course.

- If x are all exogenous, $z = x$, the above condition reads

$$\mathrm{E}(u \mid x, s) = \mathrm{E}(u \mid x) = 0.$$

  This means in order to make OLS consistent with a selected sample, we need a stronger condition $\mathrm{E}(u \mid x) = 0$ even when the sampling is exogenous.

- If s is independent of (x, y) (missing at random), then $\mathrm{E}(sxu) = \mathrm{E}(xu)$ and the usual $\mathrm{E}(ux) = 0$ guarantees the unbiasedness and consistency of OLS.

- As an application, suppose the population model is

$$y = x'\beta + u, \quad \mathrm{E}(u \mid x) = 0,$$

  and s is *correlated* with u.

- But suppose s is a deterministic function of (x, v) for a variable v. Further suppose (u, v) is independent of x.

- We have

$$\mathrm{E}(y \mid x, v) = x'\beta + \mathrm{E}(u \mid x, v) = x'\beta + \mathrm{E}(u \mid v).$$

- Suppose v has mean zero and $\mathrm{E}(u \mid v) = \gamma v$. Then

$$\mathrm{E}(y \mid x, v) = x'\beta + \gamma v.$$

  Because s is a function of (x, v), we can use OLS of y on x and v using only the selected sample to estimate consistently β and γ.

- In practice, v needs to be estimated. This idea will be used latter to treat self-selection issue.

Nonlinear models

- Suppose now
$$\mathrm{E}(y \mid x) = m(x, \beta_0),$$
for some parametric function m.

- Also assume exogenous selection,
$$\mathrm{E}(u \mid x, s) = \mathrm{E}(y - m(x, \beta_0) \mid x, s) = 0 \iff \mathrm{E}(y \mid x, s) = \mathrm{E}(y \mid x).$$

- We want to investigate how does the NLS estimator work in the selected sample. The NLS on the selected sample solves
$$\min_{\beta} n^{-1} \sum_{i=1}^{n} s_i (y_i - m(x_i, \beta))^2.$$

- As an M estimator, the consistency relies on minimizer of $\mathrm{E}\big(s_i(y_i - m(x, \beta))^2\big)$.

- We write
$$\mathrm{E}\big(s(y - m(x, \beta))^2\big) = \mathrm{E}\Big(s \cdot \mathrm{E}\big([y - m(x, \beta)]^2 \mid x, s\big)\Big).$$

- The conditional expectation $\mathrm{E}\big([y - m(x, \beta)]^2 \mid x, s\big)$ equals
$$\mathrm{E}(u^2 \mid x, s) + [m(x, \beta_0) - m(x, \beta)]^2 + 2[m(x, \beta_0) - m(x, \beta)]\, \mathrm{E}(u \mid x, s)$$
where $u \equiv y - m(x, \beta_0)$.

- By the exogenous selection condition, $\mathrm{E}(u \mid x, s) = 0$.

- The term $\mathrm{E}(u^2 \mid x, s)$ does not depend on β. The second term is minimized at $\beta = \beta_0$, though usually not uniquely for given x. (Given x, $m(x, \beta_0) - m(x, \beta) = 0$ is one equation with $\dim(\beta)$ unknowns.)

- The unconditional expectation is
$$\mathrm{E}(s[y - m(x, \beta)]^2) = \mathrm{E}(s \cdot \mathrm{E}(u^2 \mid x, s)) + \mathrm{E}\big(s \cdot [m(x, \beta_0) - m(x, \beta)]^2\big).$$
We need to assume that the second part is uniquely minimized at β_0. Note that the second part equals
$$\mathrm{E}\big((m(x, \beta_0) - m(x, \beta))^2 \mid s = 1\big) \times \Pr(s = 1).$$
We need unique minimization for the selected sample.

- MLE with the selected sample can be understood similarly since it is also M estimator.
- The log likelihood in the selected sample is

$$\sum_{i=1}^{n} s_i \ell(y_i, x_i; \theta).$$

- If the selection is exogenous in the sense that

$$F(y_i \mid x_i, s_i) = F(y_i \mid x_i),$$

then θ_0 also maximizes the expected value of the selected log likelihood $\mathrm{E}(s \cdot \ell(y, x; \theta))$.

- The argument is as follows,

$$\begin{aligned}
\mathrm{E}(s \cdot \ell(y, x; \theta)) &= \mathrm{E}(s \cdot \mathrm{E}(\ell(y, x; \theta) \mid x, s)) \\
&= \mathrm{E}(s \cdot \mathrm{E}(\ell(y, x; \theta) \mid x)) \\
&= \Pr(s = 1)\, \mathrm{E}(\mathrm{E}(\ell(y, x; \theta) \mid x) \mid s = 1).
\end{aligned}$$

We know from the Kullback-Leibler information inequality that $\mathrm{E}(\ell(y, x; \theta_0) \mid x) \geq \mathrm{E}(\ell(y, x; \theta) \mid x)$ for all θ and all x. Thus,

$$\mathrm{E}(\mathrm{E}(\ell(y, x; \theta) \mid x) \mid s = 1) \leq \mathrm{E}(\mathrm{E}(\ell(y, x; \theta_0) \mid x) \mid s = 1)$$

- We conclude that

$$\mathrm{E}(s \cdot \ell(y, x; \theta_0)) \geq \mathrm{E}(s \cdot \ell(y, x; \theta))$$

for all θ.

- For identification, we need to establish the uniqueness of maximization. This depends on the distribution of x and the nature of sample selection.

3 Selection on outcome variable

Selection on the response variable: truncated regression

- Now consider the case where the rule for sample selection depends in a known and deterministic way on the response variable.
- Assume the rule is that for *known* constants a_1 and a_2,

$$s_i = 1(a_1 < y_i < a_2).$$

If $a_1 = -\infty$ and $a_2 = \infty$, we have random sample.

- Assume a linear conditional expectation

$$\mathrm{E}(y \mid x) = x'\beta.$$

- Exogenous selection no longer holds, since $\mathrm{E}(y \mid x, s) \neq \mathrm{E}(y \mid x)$ or equivalently $\mathrm{E}(u \mid x, s) \neq 0$. OLS returns inconsistent estimator for β.

- Assume that the population conditional density is $f(y|x;\beta,\gamma)$. The density conditional on $s = 1$ is

$$\Pr(y \mid x, s = 1) = \frac{f(y|x;\beta,\gamma)}{\Pr(a_1 < y < a_2 \mid x)} = \frac{f(y|x;\beta,\gamma)}{F(a_2|x;\beta,\gamma) - F(a_1|x;\beta,\gamma)},$$

  where $F(\cdot|x;\beta,\gamma)$ is the CDF of $f(y|x;\beta,\gamma)$.

- Given the density function for the selected sample, we can use MLE. The log likelihood function for a selected sample of size n is

$$\sum_{i=1}^{n} \log f(y_i|x_i;\beta,\gamma) - \log\Big[F(a_2|x_i;\beta,\gamma) - F(a_1|x_i;\beta,\gamma)\Big]$$

- When $F(y|x) = \mathcal{N}(x'\beta, \sigma^2)$, this is often called the "truncated Tobit" or "truncated normal regression model".

- In Stata, the command for truncated *normal* regression is

```
truncreg y x1 ... xK, ll(a1) ul(a2)
```

- For other distributions, it is not difficult program the log likelihood function and run MLE in Matlab.

- *Unlike tobit model*, we can interpret the estimates of β as if we had run a linear regression using random sample.

- Easy to extend to case where the limits changes with i, that is (a_{i1}, a_{i2}). The condition becomes

$$F(y_i|x_i, a_{i1}, a_{i2}) = F(y_i|x_i).$$

4 Probit selection equation

Example of probit selection model: estimating wage equation

- To estimate the wage equation $\mathrm{E}(wage^o \mid x)$ where $wage^o$ is the wage offer, we need a random sample. But in a random sample, some will not be working and their wage offer is unobserved.

- Consider the labor supply to maximize the utility

$$\max_{h \in [0,168]} U_i(w_i^o h + a_i, h).$$

  So $w_i^o h + a_i$ is income, where a_i is non-labor income, such unemployment insurance.

- If $\partial U_i(w_i^o h + a_i, h)/\partial h|_{h=0} \leq 0$, then $h_i = 0$ is the optimum.

$$\left.\frac{\partial U_i(w_i^o h + a_i, h)}{\partial h}\right|_{h=0} \leq 0$$

- Equivalently, if

$$w_i^o \leq -\frac{u'_{i,h}(a_i, 0)}{u'_{i,y}(a_i, 0)},$$

  where $u'_{i,h}$ and $u'_{i,y}$ is the marginal disutility of working and marginal utility of income, respectively.

- Usually call the right hand side the reservation wage, w_i^r.

- Hence the person works only if

$$w_i^o > w_i^r.$$

- The trouble is that we *do not observe reservation wage* w_i^r. This is called incidental truncation.

- Model:

$$\begin{aligned} w_i^o &= \exp(x'_{i1}\beta_1 + u_{i1}) \\ w_i^r &= \exp(x'_{i2}\beta_2 + \gamma_2 a_i + u_{i2}). \end{aligned}$$

- We observe w_i^o if $\log w_i^o > \log w_i^r$ or

$$x'_{i1}\beta_1 + u_{i1} - x'_{i2}\beta_2 - \gamma_2 a_i - u_{i2} > 0.$$

- Let x include all nonredundant elements of x_{i1}, x_{i2} and a_i. So $x_1 \subset x$.

- Let $y_1 \equiv \log w_i^o$ and let y_2 be the binary labor force participation indicator (selection indicator). We have two equations

$$y_1 = x_1'\beta_1 + u_1 \qquad \text{(outcome eq)}$$
$$y_2 = 1(x'\delta_2 + v_2 > 0). \qquad \text{(selection eq)}$$

  We observe y_1 only if $y_2 = 1$.

- a_i is interesting. Note a_i is something that affects selection but not outcome. It is *excluded* from outcome function. This *excluded variable* turns out to be important for estimation! Indeed, it usually becomes indispensible for semiparametric estimation.

General model

- Notation for the general population model

$$y_1 = x_1'\beta_1 + u_1 \qquad \text{(outcome eq)}$$
$$y_2 = 1(x'\delta_2 + v_2 > 0). \qquad \text{(selection eq)}$$

- x in the selection equaton includes x_1.

- For those of x_1 that do **not** affect selection, their associated coefficients in δ_2 are zeros.

- Example:

$$y_1 = \beta_{1,1} + x_{1,2}\beta_{1,2} + x_{1,3}\beta_{1,3} + u_1$$
$$y_2 = 1(\delta_{2,1} + x_{1,2}\delta_{2,2} + x_{1,3} \cdot 0 + x_{1,4}\delta_{2,4} + v_2 > 0).$$

  Here

$$x_1 = (1, x_{1,2}, x_{1,3})' \quad x = (x_1', x_{1,4})',$$

  and $x_{1,3}$ affects outcome but not selection.

Assumption 1. *1.* (x, y_2) *are always observed,* y_1 *is observed only when* $y_2 = 1$*; 2.* (u_1, v_2) *is independent of* x *with zero mean; 3.* $v_2 \sim \mathcal{N}(0, 1)$*; 4.* $\mathrm{E}(u_1 \mid v_2) = \gamma_1 v_2$*.*

- We have to normalize the variance of v_2 for identification, since $y_2 = 1(x'\delta_2 + v_2 > 0)$ and $y_2 = 1(x'(\delta_2 c) + v_2 c > 0)$ give rise to the same distribution.

- We assume normal distribution for v_2, however, $(u_1, v_2)'$ does not have to follow bivariate normal.

- Amemiya (1985) calls it type II Tobit model, though it is a sample selection model.
- When $y_2 = 0$, do not let $y_1 = 0$. It does not make sense and unnecessary.
- Since y_1 is observed only when $y_2 = 1$, what we can hope to estimate is $\mathrm{E}(y_1|x, y_2 = 1)$. How does $\mathrm{E}(y_1 \mid x, y_2 = 1)$ relate β_1?
- It is helpful to write

$$y_1 = x_1'\beta_1 + \mathrm{E}(u_1 \mid x, y_2 = 1) + e \tag{2}$$

$$e = u_1 - \mathrm{E}(u_1 \mid x, y_2 = 1) \tag{3}$$

- Obviously, $\mathrm{E}(e \mid x, y_2 = 1) = 0$. Since x_1 is at least a part of x, $\mathrm{E}(e \mid x_1, y_2 = 1) = 0$.
- If we know $g(x) \equiv \mathrm{E}(u_1 \mid x, y_2 = 1)$, we can estimate β_1 by OLS of y_1 on x_1 and $g(x)$ using only the selected sample!
- The aim is to get $\mathrm{E}(u_1 \mid x, y_2 = 1)$.
- We write

$$\begin{aligned}
\mathrm{E}(u_1 \mid x, y_2 = 1) &= \mathrm{E}(u_1 \mid x, v_2 > -x'\delta_2) \\
&= \mathrm{E}(\mathrm{E}(u_1 \mid x, v_2) \mid x, v_2 > -x'\delta_2) \\
&= \mathrm{E}(\gamma_1 v_2 \mid x, v_2 > -x'\delta_2) \\
&= \gamma_1 \lambda(x'\delta_2),
\end{aligned} \tag{4}$$

  where $\lambda(\cdot)$ is the inverse Mills ratio.
- We did not assume that (u_1, v_2) follows bivariate normal. We just need $\mathrm{E}(u_1 \mid v_2) = \gamma_1 v_2$.
- If $\gamma_1 = 0$,

$$\mathrm{E}(u_1 \mid x, y_2 = 1) = 0.$$

  Thus,

$$\mathrm{E}(y_1 \mid x, y_2 = 1) = x'\beta_1 = \mathrm{E}(y \mid x),$$

  and OLS can consistently estimate β_1 using the selected sample.
- When $\gamma_1 = 0$, sample selection is not a problem!
- From eq. (2), eq. (3) and eq. (4), we conclude

$$y_1 = x_1'\beta_1 + \gamma_1\lambda(x'\delta_2) + e, \quad \mathrm{E}(e \mid x, y_2 = 1) = 0.$$

- It is clear that an OLS of y_1 on x_1 using the selected sample omits the term $\lambda(x'\delta_2)$ generally leads to inconsistent estimation of β_1.
- The bias is usually called "selection bias".
- Essentially, the bias is due to omitted variable.

Heckman's two step estimation procedure

- Here is the two procedure:
  1. Obtain the probit estimate $\hat{\delta}_2$ from the probit model
  $$\Pr(y_2 = 1 \mid x_i) = \Phi(x_i'\delta_2),$$
  using all N observations. Then obtain the estimated inverse Mills ratio $\hat{\lambda}_{i2} \equiv \lambda(x_i'\hat{\delta}_2)$ for those whose $y_{i2} = 1$ (in selected sample).
  2. Obtain $\hat{\beta}_1$ and $\hat{\gamma}_1$ from the OLS regression on the selected sample,
  $$y_{i1} \text{ on } x_{i1}, \hat{\lambda}_{i2}, \quad \text{for } i \text{ such that } y_{i2} = 1.$$
  These estimators are consistent and $\sqrt{N}$ asymptotically normal.
- This procedure is sometimes called Heckit due to the corresponding stata command `heckman`.
- The standard t test for $H_0 : \gamma_1 = 0$ can be executed to test selection bias.
- Technically, x_1 does not need to be a subset of x.
- *However*, if $x_1 = x$, the second step estimation will suffer from severe multicollinearity problem, because the inverse Mills ratio $\lambda(x_i'\hat{\delta}_2)$ is almost linear in $x_i'\hat{\delta}_2$. *It is very helpful to have excluded variables!* (Of course the coefficient in front of the excluded variable should be nonzero.)
- The variance needs careful adjustment due to two-step feature.

Example: estimate married women's wage offer equation

```
. use mroz

. des lwage inlf nwifeinc
```

```
              storage  display     value
variable name   type   format      label      variable label
-------------------------------------------------------------------------------
lwage           float  %9.0g                  log(wage)
inlf            byte   %9.0g                  =1 if in lab frce, 1975
nwifeinc        float  %9.0g                  (faminc - wage*hours)/1000
. sum lwage inlf educ kidslt6 nwifeinc

    Variable |       Obs        Mean    Std. Dev.       Min        Max
-------------+--------------------------------------------------------
       lwage |       428    1.190173    .7231978  -2.054164   3.218876
        inlf |       753    .5683931    .4956295          0          1
        educ |       753    12.28685    2.280246          5         17
     kidslt6 |       753    .2377158     .523959          0          3
    nwifeinc |       753    20.12896     11.6348  -.0290575         96

. reg lwage educ exper expersq

      Source |       SS       df       MS              Number of obs =     428
-------------+------------------------------           F(  3,   424) =   26.29
       Model |  35.0222967     3  11.6740989           Prob > F      =  0.0000
    Residual |  188.305144   424  .444115906           R-squared     =  0.1568
-------------+------------------------------           Adj R-squared =  0.1509
       Total |  223.327441   427  .523015084           Root MSE      =  .66642

------------------------------------------------------------------------------
       lwage |      Coef.   Std. Err.      t    P>|t|     [95% Conf. Interval]
-------------+----------------------------------------------------------------
        educ |   .1074896   .0141465     7.60   0.000     .0796837    .1352956
       exper |   .0415665   .0131752     3.15   0.002     .0156697    .0674633
     expersq |  -.0008112   .0003932    -2.06   0.040    -.0015841   -.0000382
       _cons |  -.5220406   .1986321    -2.63   0.009    -.9124667   -.1316144
------------------------------------------------------------------------------

. heckman lwage educ exper expersq, select(inlf = educ exper expersq nwifeinc
          age kidslt6 kidsge6) twostep

Heckman selection model -- two-step estimates   Number of obs      =       753
(regression model with sample selection)        Censored obs       =       325
                                                Uncensored obs     =       428

                                                Wald chi2(6)       =    180.10
                                                Prob > chi2        =    0.0000

------------------------------------------------------------------------------
             |      Coef.   Std. Err.      z    P>|z|     [95% Conf. Interval]
-------------+----------------------------------------------------------------
lwage        |
        educ |   .1090655    .015523     7.03   0.000     .0786411      .13949
       exper |   .0438873   .0162611     2.70   0.007     .0120163    .0757584
     expersq |  -.0008591   .0004389    -1.96   0.050    -.0017194    1.15e-06
       _cons |  -.5781032   .3050062    -1.90   0.058    -1.175904     .019698
-------------+----------------------------------------------------------------
```

```
inlf         |
        educ |   .1309047   .0252542     5.18   0.000     .0814074     .180402
       exper |   .1233476   .0187164     6.59   0.000     .0866641    .1600311
     expersq |  -.0018871      .0006    -3.15   0.002     -.003063   -.0007111
    nwifeinc |  -.0120237   .0048398    -2.48   0.013    -.0215096   -.0025378
         age |  -.0528527   .0084772    -6.23   0.000    -.0694678   -.0362376
     kidslt6 |  -.8683285   .1185223    -7.33   0.000    -1.100628    -.636029
     kidsge6 |    .036005   .0434768     0.83   0.408     -.049208    .1212179
       _cons |   .2700768    .508593     0.53   0.595    -.7267473    1.266901
-------------+----------------------------------------------------------------
mills        |
      lambda |   .0322619   .1336246     0.24   0.809    -.2296376    .2941613
-------------+----------------------------------------------------------------
         rho |    0.04861
       sigma |  .66362875
      lambda |  .03226186   .1336246
------------------------------------------------------------------------------

. heckman lwage educ exper expersq, select(inlf =educ exper expersq nwifeinc age
          kidslt6 kidsge6)

Heckman selection model                         Number of obs      =       753
(regression model with sample selection)        Censored obs       =       325
                                                Uncensored obs     =       428

                                                Wald chi2(3)       =     59.67
Log likelihood = -832.8851                      Prob > chi2        =    0.0000

------------------------------------------------------------------------------
             |      Coef.   Std. Err.      z    P>|z|     [95% Conf. Interval]
-------------+----------------------------------------------------------------
lwage        |
        educ |   .1083502   .0148607     7.29   0.000     .0792238    .1374767
       exper |   .0428369   .0148785     2.88   0.004     .0136755    .0719983
     expersq |  -.0008374   .0004175    -2.01   0.045    -.0016556   -.0000192
       _cons |  -.5526973   .2603784    -2.12   0.034     -1.06303   -.0423651
-------------+----------------------------------------------------------------

inlf         |
        educ |   .1313415   .0253823     5.17   0.000     .0815931    .1810899
       exper |   .1232818   .0187242     6.58   0.000     .0865831    .1599806
     expersq |  -.0018863   .0006004    -3.14   0.002     -.003063   -.0007095
    nwifeinc |  -.0121321   .0048767    -2.49   0.013    -.0216903    -.002574
         age |  -.0528287   .0084792    -6.23   0.000    -.0694476   -.0362098
     kidslt6 |  -.8673988   .1186509    -7.31   0.000     -1.09995   -.6348472
     kidsge6 |   .0358723   .0434753     0.83   0.409    -.0493377    .1210824
       _cons |   .2664491   .5089578     0.52   0.601    -.7310898    1.263988
-------------+----------------------------------------------------------------
     /athrho |    .026614    .147182     0.18   0.857    -.2618573    .3150854
    /lnsigma |  -.4103809   .0342291   -11.99   0.000    -.4774687   -.3432931
-------------+----------------------------------------------------------------
         rho |   .0266078   .1470778                     -.2560319    .3050564
       sigma |   .6633975   .0227075                      .6203517    .7094303
      lambda |   .0176515   .0976057                     -.1736521    .2089552
```

```
------------------------------------------------------------------------------
LR test of indep. eqns. (rho = 0):   chi2(1) =     0.03   Prob > chi2 = 0.8577
------------------------------------------------------------------------------
```

Endogenous explanatory variables

- Consider the model

$$
\begin{aligned}
y_1 &= z_1\delta_1 + \alpha_1 y_2 + u_1 \\
y_2 &= z_2\delta_2 + v_2 \\
y_3 &= 1(z'\delta_3 + v_3 > 0).
\end{aligned}
$$

  Here y_2 is endogenous variable, and y_3 is the selection indicator.

- z includes all nonredundant variables in z_1 and z_2. For those z in z_1 and z_2 that does not affect selection (y_3), their associated δ coefficients are zero.

- y_{2i} can be either always observable or observed only if $y_{3i} = 1$.

- If there was no selection issues (y_1, y_2 and z are always observable), 2SLS IV is enough. We will show that 2SLS *with the inverse Mills ratio added* to the regressors is consistent.

Assumption 2. *1.* (z, y_3) *is always observed,* (y_1, y_2) *is observed when* $y_3 = 1$ *2.* $(u_1, v_3) \perp\!\!\!\perp z$*; 3.* $v_3 \sim \mathcal{N}(0,1)$*; 4.* $\mathrm{E}(u_1 \mid v_3) = \gamma_1 v_3$*; 5.* $\mathrm{E}(z_2 v_2) = 0$ *and let* $z_2'\delta_2 = z_1\delta_{21} + z_{21}'\delta_{22}$ *with* $\delta_{22} \neq 0$.

- Item 5 requires at least one IV for y_2.

- The relationship between z_2 and z is subtle. For now, let's say this assumption allows for $z_2 = z$.

- We write

$$
y_1 = z_1'\delta_1 + \alpha_1 y_2 + g(z) + e_1, \tag{5}
$$

  where

$$
g(z) = \mathrm{E}(u_1 \mid z, y_3 = 1), \quad e_1 = u_1 - \mathrm{E}(u_1 \mid z, y_3 = 1).
$$

  Obviously, $\mathrm{E}(e_1 \mid z, y_3 = 1) = 0$.

- Hence provided that we know $g(z)$, eq. (5) using only selected sample is the usual linear regression with one endogenous variable y_2 (because v_2 might be correlated with u_1). 2SLS can be applied to estimate δ_1 and α_1.

- By the previous results, we know that

$$g(z) = \mathrm{E}(u_1 \mid z, y_3 = 1) = \gamma_1 \lambda(z'\delta_3).$$

- Here is two step procedure for the case with endogenous explanatory variable:

  1. Obtain $\hat{\delta}_3$ from probit of y_3 on z using all observations. Obtain the estimated inverse Mills ratio, $\hat{\lambda}_{i3} = \lambda(z'\hat{\delta}_3)$.
  2. Using the selected subsample (for which we observe both y_1 and y_2), estimate the equation

$$y_{i1} = z_{i1}'\delta_1 + \alpha_1 y_{i2} + \gamma_1 \hat{\lambda}_{i3} + \varepsilon_i,$$

  by 2SLS, using instruments $(z_{i2}, \hat{\lambda}_{i3})$.

- Now it is clear that z_2 in $y_2 = z_2'\delta_2 + v_2$ should contain an instrument z_{22} that does not belong to z_1. Otherwise the rank condition of the 2SLS will fail.

- It is OK that $z = z_2$, that is there is no variable that affects only selection but not endogenous y_2. The consequence though is that the reduced form for y_2 will suffer from collinearity because inverse Mills ratio is almost linear is $z'\delta_3$.

- If $z = z_2$, we still have excluded variable that affects selection but not the outcome *directly*. That excluded variable is $z_{2,2}$, the instrumental variable for y_2.

Roy model

- The selection is based on y_3,

$$y_3 = 1(x'\delta_3 + v_3 > 0).$$

  If $y_3 = 1$, we observe y_1 and

$$y_1 = x_1'\beta_1 + u_1.$$

  If $y_3 = 0$, we observe y_2 and

$$y_2 = x_2'\beta_2 + u_2.$$

- Immigration (US-Mexico) and migration (urban-rural) are good examples.

Assumption 3. *1.* (x, y_3) *are always observed,* y_1 *is observed when* $y_3 = 1$ *and* y_2 *is observed when* $y_3 = 0$*; 2.* (u_1, u_2, v_3) *is independent of* x *with zero mean; 3.* $v_3 \sim \mathcal{N}(0, 1)$*; 4.* $\mathrm{E}(u_1 \mid v_3) = \gamma_1 v_3$*; 5.* $\mathrm{E}(u_2 \mid v_3) = \gamma_2 v_3$*.*

- For $k = 1, 2$, we then have

$$y_k = x_k'\beta_k + g(x) + e_k, \quad g_k(x) = \mathrm{E}(u_k \mid x, y_3 = 1) \quad e_k = u_k - g(x).$$

  From the previous results, we have known that $g_k(x) = \gamma_k \lambda(x'\delta_3)$.

- It is obvious that we can apply the two step procedure to estimate β_1, β_2, γ_1 and γ_2.

5 Semiparametric sample selection model

The role of propensity score

- Consider a more general model,

$$\begin{aligned} y_1 &= x_1'\beta_1 + u_1 \\ y_2 &= 1(\delta(x) - v_2 > 0). \end{aligned}$$

  We replaced $x'\delta_2$ with a general function $\delta(x)$ of x. "$-v_2$" is not a typo.

- We know that

$$\mathrm{E}(y_1 \mid x, y_2 = 1) = x_1'\beta_1 + \mathrm{E}(u_1 \mid x, y_2 = 1).$$

- The point of Heckman and Robb (1985, *Journal of Econometrics*) is that $\mathrm{E}(u_1 \,|\, x, y_2 = 1)$ is a function of $\Pr(y_2 = 1 \mid x)$, which is called propensity score in the literature.

- To see this point, we have

$$\begin{aligned} \mathrm{E}(u_1 \mid x, y_2 = 1) &= \mathrm{E}(u_1 \mid x, v_2 < \delta(x)) \\ &= \frac{\int_{-\infty}^{\infty} \int_{-\infty}^{\delta(x)} u_1 f(u_1, v_2|x) \,\mathrm{d}\, v_2 \,\mathrm{d}\, u_1}{\int_{-\infty}^{\infty} \int_{-\infty}^{\delta(x)} f(u_1, v_2|x) \,\mathrm{d}\, v_2 \,\mathrm{d}\, u_1}. \end{aligned} \tag{6}$$

- Assume $(u_1, v_2) \perp\!\!\!\perp x$ and the CDF of v_2 is strictly increasing. We have

$$p(x) \equiv \Pr(y_2 = 1 \mid x) = F_{v_2}(\delta(x)) \Rightarrow \delta(x) = F_{v_2}^{-1}(\Pr(y_2 = 1 \mid x)).$$

- We have

$$\mathrm{E}(u_1 \mid x, y_2 = 1) = \frac{\int_{-\infty}^{\infty} \int_{-\infty}^{F_{v_2}^{-1}(p(x))} u_1 f(u_1, v_2) \,\mathrm{d}\, v_2 \,\mathrm{d}\, u_1}{\int_{-\infty}^{\infty} \int_{-\infty}^{F_{v_2}^{-1}(p(x))} f(u_1, v_2) \,\mathrm{d}\, v_2 \,\mathrm{d}\, u_1}.$$

- It is clear now that $\mathrm{E}(u_1 \mid x, y_2 = 1)$ is a function of $p(x)$.

Two-step estimation method by Heckman and Robb (1985)

- Here is the procedure:

  1. estimate propensity score $\hat{p}$ using a parametric, a semiparametric or a nonparametric method;
  2. estimate
  $$y_{1i} = x_{1i}'\beta_1 + g(\hat{p}(x_i)) + \varepsilon_i.$$
  $g(\hat{p}(x_i))$ can be written as a series expansion, such as power series,
  $$g(p) = \alpha_0 + \alpha_1 p + \alpha_2 p^2 + \cdots + \alpha_K p^K.$$
  See chapter 8 of Wasserman (2006, *All of Nonparametric Statistics*).

- When $g(p)$ is estimated nonparametrically, the *intercept* term in $x_{1i}'\beta_1$ is not separately identified.

- If $x = x_1$ (no excluded variable), $x_{1i}'\beta_1$ and $g(p)$ are not separately identified.

The Robinson (1988, *Econometrica*) estimator for the partially linear model

- The previous problem fits the partially linear model
$$y_{1i} = x_{1i}'\beta_1 + g(p_i) + \varepsilon_i. \tag{7}$$

- The idea is as follows,
$$\mathrm{E}(y_i \mid p_i) = \mathrm{E}(x_{1i}' \mid p_i)\beta_1 + g(p_i) + \mathrm{E}(\varepsilon_i \mid p_i). \tag{8}$$
  Equation (7) minus eq. (8) leads
$$y_{1i} - \mathrm{E}(y_{1i} \mid p_i) = (x_{1i} - \mathrm{E}(x_{1i} \mid p_i))'\beta_1 + (\varepsilon_i - \mathrm{E}(\varepsilon_i \mid p_i)).$$

- Note all conditional expectations can be estimated nonparametrically.

- Robinson showed that $\hat{\beta}_1$ converge at the rate of $n^{-1/2}$. The asymptotic variance is complicated.

- Stata command `semipar`

Ahn and Powell (1993, *Journal of Econometrics*)

- Model:

$$y_{1i} = x_{1i}'\beta_1 + u_{1i}$$
$$y_{2i} = 1(x_i'\delta_2 + v_{2i} > 0).$$

- By eq. (6), we know

$$y_{1i} = x_{1i}'\beta_1 + \varphi(x_i'\delta_2) + \varepsilon_i,$$

  where $\varepsilon_i = y_{1i} - \mathrm{E}(y_{1i} \mid x_i, y_{2i} = 1)$.

- The idea of Ahn-Powell estimator is to find individuals with close index $x_i'\delta_2$. For those individuals,

$$\varphi(x_i'\delta_2) \approx \varphi(x_i'\delta_2).$$

- Their estimator is to minimize

$$\min_b \sum_{i=1}^n \sum_{j=1}^n (y_i - y_j - (x_{1i} - x_{ji})'\beta_1)^2 K((x_i'\hat{\delta}_2 - x_j'\hat{\delta}_2)/h_n).$$